مدارس الإمارات الوطنية
Emirates National Schools

Grade: 6 | Term 1, Week 11 (9 Nov to 13 Nov) | TCD: Future Skills 6: Applied Technology Foundations | Session length: 2 hrs

T1 WEEK: 11	**Lesson 1: Unimat Cool Tools and 3D Print**
Standard(s):	Science: properties of materials; Technology: safe use of machine tools, CAD to 3D print workflow; Math: measurement and precision
Literacy Strategies:	Machine tool picture cards; safety pledge read aloud; frame: I used the ... to ... safely by ...
Assessment:	Finished maker mark checked against the quality checklist; tool safety observation; exported STL; live quiz on the TCD website (counts towards the My Maker Mark certificate).
Lesson Objectives:	We are learning to use small machine tools safely to finish a product to a high quality, and to prepare a model for 3D printing.
Success Criteria:	I can name the Unimat Cool Tools machines and what each one does. I can sand and finish my maker mark safely using the sanding machine. I can follow the safety rules for machine tools every time. I can check and export my Junior Engineer model for printing.
Targeted Vocabulary	Unimat, sanding machine, jigsaw, lathe, drill, finish, grit, STL, quality
Resources	Unimat Cool Tools kit (sanding machine, hand drill, jigsaw) with 14 V safety transformer, Safety goggles, Sandpaper (120 and 240 grit), Laser-cut maker marks from Week 10, Keyrings and pliers, Computers with Tinkercad, 3D printer and PLA filament Accompanying resource: Grade6_Week11_Presentation.pptx (slides, hook video and live quiz code on the TCD website), Grade6_Week11_Worksheet and Grade6_Week11_Lesson_Plan.
National Identity Integration	Link to Emirati craftspeople such as dhow builders, who shape and finish wood by hand with great care, and to the UAE's growing advanced manufacturing sector.
AI/Digital Learning	Tinkercad; slicer on the teacher's computer; hook video; live quiz on the TCD website.
Direct Instruction: I Do	Introduce the Unimat Cool Tools (slide 6). Teach machine tool safety; students read the safety pledge aloud (slides 7 and 11). Demonstrate sanding and finishing (slide 8).
Guided Practice: We Do	Students practise on an offcut at the sanding machine one at a time while the teacher watches. Class reviews the print-ready checklist for the Junior Engineer model (slide 9).
Independent Practice: You Do	Rotation: half the class finishes maker marks at the machines while the other half checks and exports their Junior Engineer models; then swap (slides 12 and 10). Challenge levels on the board (slide 13). Live quiz on tablets (slide 14).
Differentiated Instruction: Beginning / On / Above	**Beginning: Sand by hand with sandpaper first, then use the sanding machine with the teacher beside you.** **On: Finish your piece on the sanding machine and export your model independently.** **Above: Use the hand drill to add a decorative detail, and write a short quality report comparing your piece with a professional product.**
Closure/Reflection	Maker mark showcase: each student shows one detail they finished. Exit ticket (slide 15). The teacher prints Junior Engineer models this week.
Inclusion / EAL Support	☑ Visual Supports ☑ Chunked Instructions ☐ Reduced Language Load ☑ Model / Worked Example ☐ Pre-teach Vocabulary ☑ TA/LSA Support ☐ Other: ☐ Extra Processing Time ☐ Text Read by TA/LSA ☐ Flexible Grouping ☐ Assistive Technology ☐ Movement / Sensory Break ☐ Scaffold / Sentence Starters

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G&T / Extension	☑ Extension / Challenge Task ☑ Real-World Application ☐ Higher-Order Questioning ☐ Curriculum Compacting ☑ Depth & Complexity Task ☐ Above-Grade-Level Texts ☐ Independent Inquiry ☐ Student-Led /Passion project ☐ Leadership / Peer Coaching ☐ Open-Ended Task ☐ Other: _____